

➤ Unit I: Introduction to Computer Networks and Physical Layer

• Introduction to Computer Networks

What is a Computer Network?

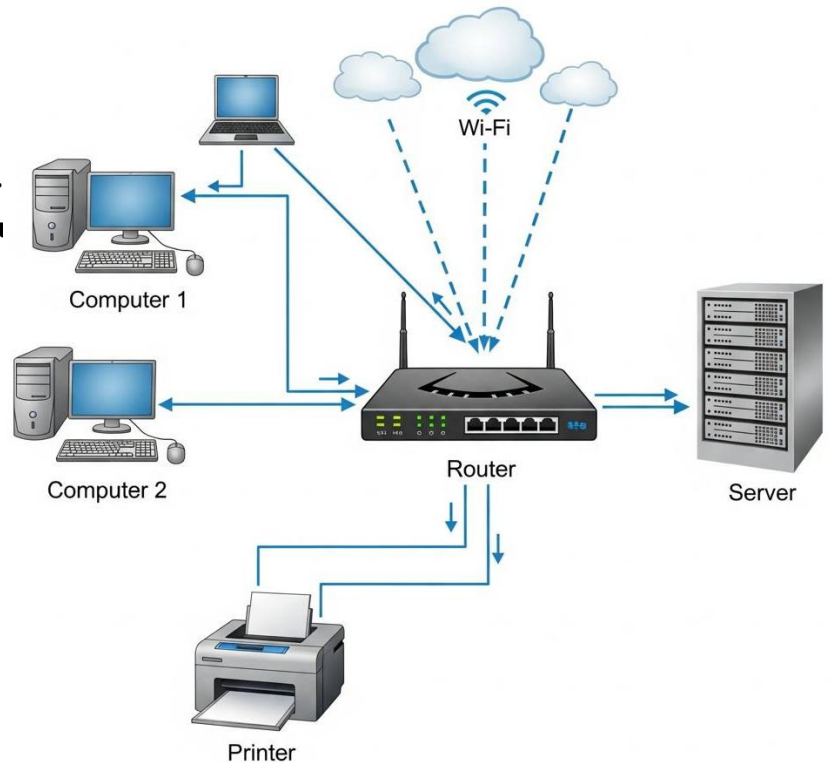
A **Computer Network** is a collection of **two or more** computers (or other devices) that **communicate, exchange data, and share resources** (e.g., files, printers, Internet connection).

A network may be connected using:

- Wired cables (Ethernet, Optical Fiber)
- Wireless technology (Wi-Fi, Bluetooth, Cellular Network)

Simple Definition (Exam View)

A **computer network** is a group of **interconnected computers and devices** that **communicate with each other** to **share data, hardware, software, and Internet resources**.



Real-Life Example

Imagine a **college computer laboratory**.

There are:

- 50 computers
- 1 printer
- 1 Internet connection
- 1 server

Instead of buying **50 printers** and **50 Internet connections**, all computers are connected through a network.

Students can:

- Print from any computer
- Access the Internet
- Share files
- Submit assignments

This is a **computer network**.

Another Example

WhatsApp

When you send a message:

1. Your phone sends data.
2. Data travels through the Internet.
3. WhatsApp server receives it.
4. Receiver gets the message.

Without computer networks, WhatsApp would not work.

- **Components of Computer Network**

A computer network consists of:

1. Computer (Node)

Devices that send or receive data.

Examples:

- Laptop
- Desktop
- Smartphone

2. Server

A powerful computer that provides services to other computers.

Example:

- College Result Server
- Google Server

3. Client

A computer that requests services from a server.

Example:

Your laptop opening Gmail.

4. Transmission Media

Medium through which data travels.

Examples:

- Ethernet Cable
- Optical Fiber
- Wi-Fi

5. Network Devices

Examples:

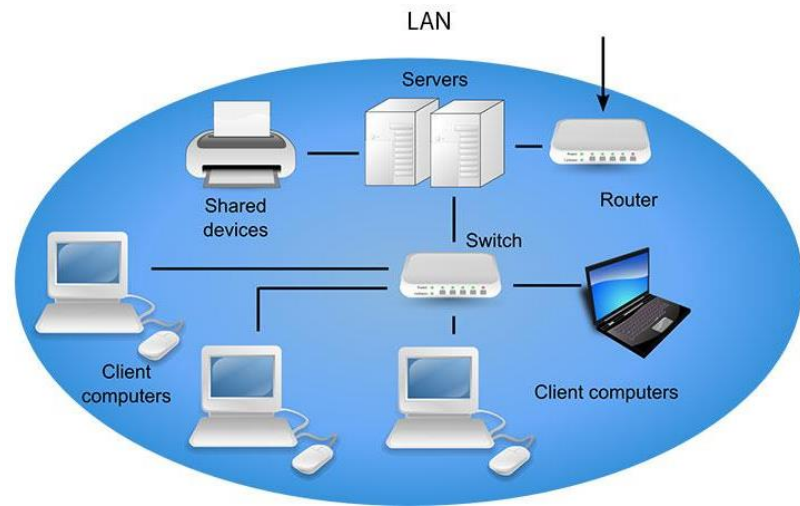
- Router
- Switch
- Hub
- Modem
- Access Point

6. Network Software

Software used for communication.

Examples:

- Windows Networking
- TCP/IP Protocol



Characteristics of Computer Network

- High speed communication
 - Resource sharing
 - Reliability
 - Scalability
 - Security
 - Remote access
-

Advantages of Computer Networks

Advantage	Explanation
Resource Sharing	Share printers, Internet and files
Fast Communication	Instant messaging and email
Cost Saving	One printer or Internet for many users
Centralized Data	Store data in one server
Easy Backup	Data backup becomes simple
Remote Access	Access files from anywhere
Collaboration	Teams work together easily

Example

A company has **100 employees**.

Instead of:

- 100 printers
- 100 Internet connections

They use:

- 2 printers
- 1 Internet connection

This saves money.

Disadvantages of Computer Networks

Disadvantage	Explanation
Security Risk	Hackers may steal data
Virus Spread	Malware spreads quickly
Network Failure	Entire work stops if network fails
Maintenance Cost	Requires skilled professionals
Data Theft	Sensitive information may be stolen

Uses of Computer Networks

Computer networks are mainly used in:

1. Business Applications
 2. Home Applications
 3. Mobile Users
-

1. Business Applications

Businesses use computer networks to improve productivity and communication.

Applications

File Sharing

Employees access the same files.
Example:
Google Drive in a company.

Resource Sharing

Share:

- Printers
- Internet
- Software

Example:

Office printer used by all employees.

Communication

Employees communicate through:

- Email
- Video conferencing
- Chat

Example:

Microsoft Teams or Zoom meetings.

Online Banking

Banks connect all branches.

Example:

When money is transferred from one city to another instantly.

Inventory Management

Stores update product information automatically.

Example:

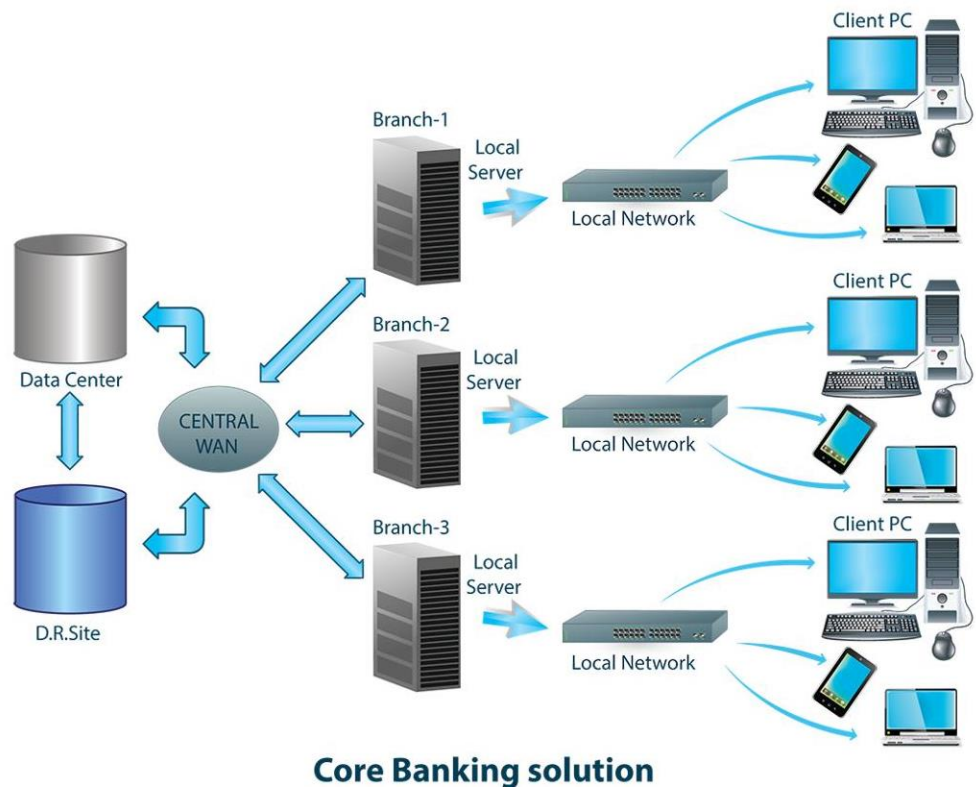
Supermarket billing system.

E-Commerce

Websites:

- Amazon
- Flipkart

Customers order products online.



Advantages

- Faster communication
- Reduced cost
- Better security
- Increased productivity
- Easy collaboration

Disadvantages

- Cyber attacks
- Expensive setup
- Maintenance required
- Internet dependency

Real-Life Example

A company has offices in:

- Pune
- Mumbai
- Delhi

Employees work on the same project through a network without travelling.

2. Home Applications

Nowadays almost every home has a computer network.

Applications

Internet Browsing

Search information using Google.

Online Education

Attend classes on:

- Google Meet
- Zoom

Entertainment

Watch:

- YouTube
- Netflix

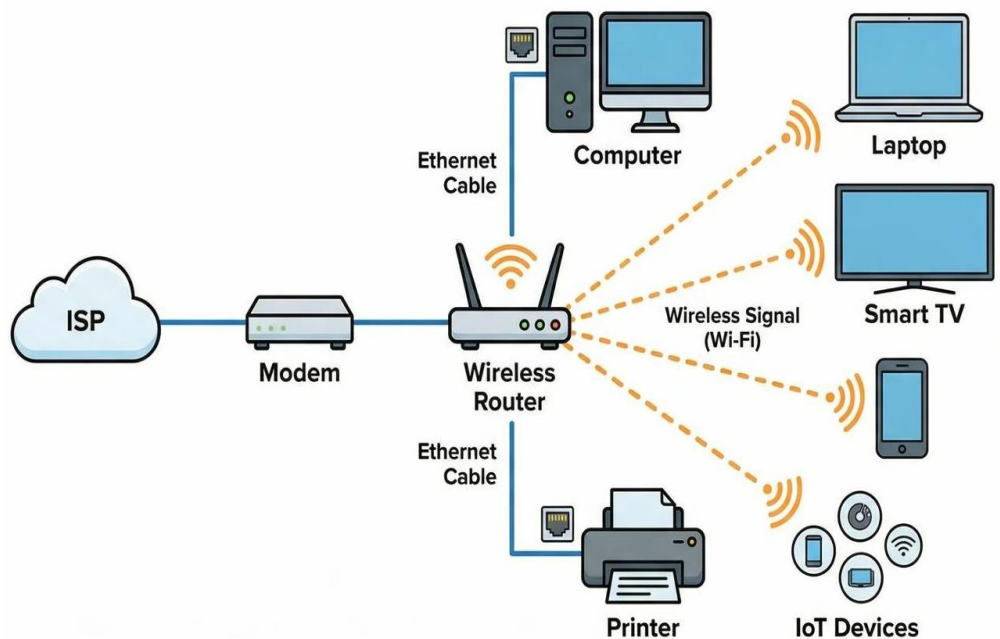
Online Shopping

Buy products from home.

Social Media

Use:

- Instagram
- Facebook
- WhatsApp



Smart Home

Control:

- Smart lights
- Smart cameras
- Smart TV
- Smart speakers

Advantages

- Entertainment
- Learning
- Communication
- Online shopping
- Smart home automation

Disadvantages

- Internet addiction
- Privacy issues
- Cybercrime
- Fake websites

Real-Life Example

A family has:

- 2 smartphones
- 1 Smart TV
- 1 Laptop
- 1 Tablet

All devices connect to one Wi-Fi router.

Everyone uses the Internet simultaneously.

3. Mobile Users

A mobile user accesses computer networks using smartphones or tablets.

Applications

Mobile Banking

Transfer money using banking apps.

GPS Navigation

Google Maps provides live location.

Video Calling

Apps:

- WhatsApp
- Google Meet

Online Shopping

Purchase products anytime.

Cloud Storage

Access files using:

- Google Drive
- OneDrive

Online Payment

Use:

- UPI
- PhonePe
- Google Pay

Advantages

- Access anywhere
- Instant communication
- Portable
- Real-time information
- Convenient

Disadvantages

- Battery drain
- Network coverage issues
- Data privacy risks
- Mobile theft
- Internet dependency

Real-Life Example

While travelling by train:

- Book tickets online.
- Use Google Maps.
- Make UPI payments.
- Attend an online meeting.

All of these are possible because of computer networks.

➤ Network Hardware (Types of Computer Networks)

Network hardware can be classified according to the **geographical area** it covers.

The five main types are:

1. **PAN (Personal Area Network)**
2. **LAN (Local Area Network)**
3. **MAN (Metropolitan Area Network)**
4. **WAN (Wide Area Network)**
5. **Internetwork (Network of Networks)**

1. PAN (Personal Area Network)

Definition

A **Personal Area Network (PAN)** is the smallest type of computer network. It connects devices belonging to **one person** within a very short distance (about **1–10 meters**).

PAN is mainly used for **personal communication** between electronic devices.

Exam Definition

PAN is a network that connects personal electronic devices within a short distance, usually up to 10 meters.

Devices Connected in PAN

- Smartphone
- Laptop
- Smartwatch
- Bluetooth Headphones
- Wireless Keyboard
- Wireless Mouse
- Fitness Band



Working of PAN

1. A user enables Bluetooth or Wi-Fi.
2. Devices discover each other.
3. Devices pair securely.
4. Data is exchanged wirelessly.

Example

You connect:

- Smartphone → Bluetooth Earbuds
- Smartphone → Smartwatch
- Laptop → Wireless Mouse

All these devices form a **PAN**.

Real-Life Example

A student is studying.

He connects:

- Laptop
- Bluetooth Mouse
- Bluetooth Keyboard
- Bluetooth Speaker

All devices communicate within a few meters.

This is a **Personal Area Network**.

Advantages

- Easy to install
- Low cost
- Wireless communication
- Portable
- Low power consumption

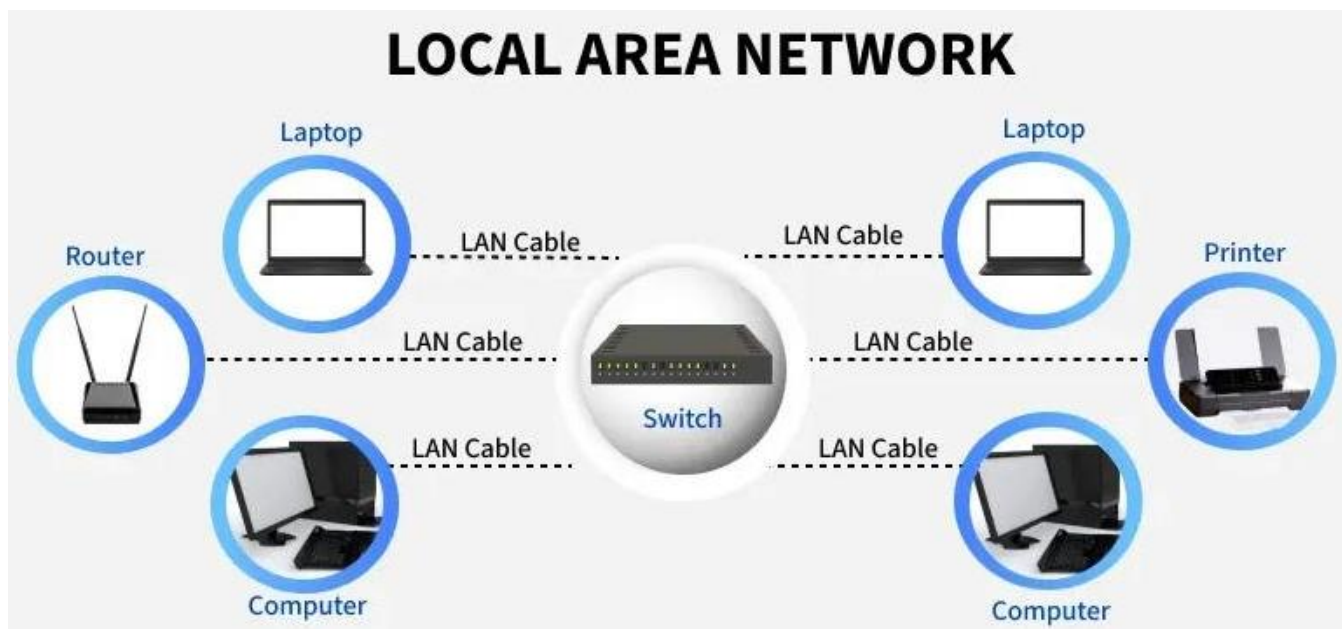
Disadvantages

- Very short range
- Low speed compared to LAN
- Limited number of devices
- Signal interference

Applications

- Bluetooth Headphones
- Smartwatch
- Fitness Tracker
- Mobile Hotspot
- Wireless Printer

2. LAN (Local Area Network)



Definition

A **Local Area Network (LAN)** connects computers within a **small geographical area**, such as:

- Home
- Office
- School
- College
- Laboratory

LAN usually covers a few meters to a few kilometers.

Exam Definition

LAN is a computer network that connects devices within a limited geographical area such as a home, office, school, or building.

Components

- Computers
- Switch
- Router
- Printer
- Server
- Ethernet Cable
- Wi-Fi Access Point

Working of LAN

1. Computers connect to a switch or Wi-Fi router.
2. Every computer receives a unique IP address.
3. Users send requests.
4. Data travels through the switch/router.
5. Destination computer receives the data.

Real-Life Example

A college computer lab has:

- 60 computers
- One printer
- One Internet connection
- One server

All computers share:

- Printer
- Internet
- Files

This is a **LAN**.

Advantages

- High speed (100 Mbps to several Gbps)
- Easy resource sharing
- Low communication cost
- Fast file transfer
- Easy maintenance
- Secure within the organization

Disadvantages

- Covers a small area only
- Initial installation cost
- Network failure affects all users
- Requires administration

Applications

- School computer labs
- Office networks
- Hospitals
- Libraries
- Banks (inside one branch)

3. MAN (Metropolitan Area Network)

Definition

A **Metropolitan Area Network (MAN)**

connects multiple LANs across a **city or metropolitan area**.

Coverage is approximately **5 km to 50 km** (or more).

Exam Definition

MAN is a network that interconnects multiple LANs over a city or metropolitan region.



Working of MAN

1. Multiple LANs are created.
2. High-speed fiber optic cables connect them.
3. Routers exchange data between LANs.
4. Users communicate across the city.

Real-Life Example

A university has:

- Main Campus
- Engineering Campus
- Medical Campus
- Hostel

All campuses are connected through fiber optic cables.

This is a **MAN**.

Another Example

A bank has **20 branches** in Pune.

All branches share customer records instantly.

This network is a **MAN**.

Advantages

- Covers an entire city
- High-speed communication
- Resource sharing between branches
- Better Internet connectivity
- Reliable communication

Disadvantages

- Expensive installation
- Difficult maintenance
- Complex management
- Security challenges

Applications

- City-wide bank branches
- Government offices
- University campuses
- Cable TV networks
- Smart city infrastructure

4. WAN (Wide Area Network)

Definition

A **Wide Area Network (WAN)** connects computers over **very large geographical areas**, such as countries or continents.

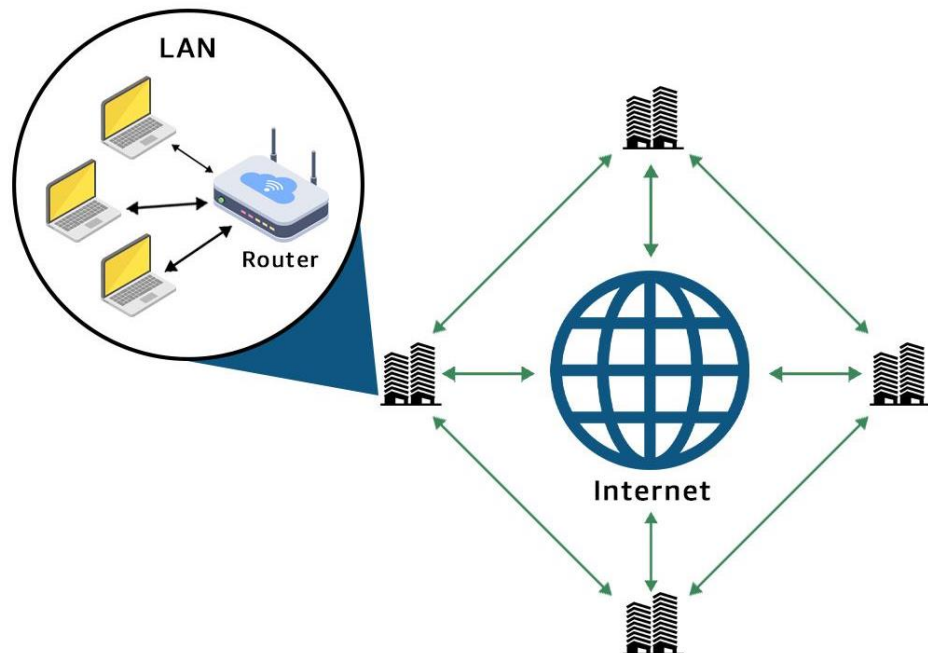
It connects multiple LANs and MANs using leased lines, satellites, fiber optics, or cellular networks.

Exam Definition

WAN is a computer network that connects multiple LANs and MANs over large geographical distances such as countries or continents.

Working of WAN

1. Each office has its own LAN.
2. Routers connect the LANs to Internet Service Providers.
3. Fiber optics, satellites, or submarine cables carry the data.
4. Destination office receives the information.



Real-Life Example

A multinational company has offices in:

- Pune
- Delhi
- London
- New York

Employees share files and attend video meetings over the WAN.

Best Example

The **Internet** is the world's largest WAN.

Advantages

- Worldwide communication
- Remote working
- Cloud access
- Global resource sharing
- Supports international business

Disadvantages

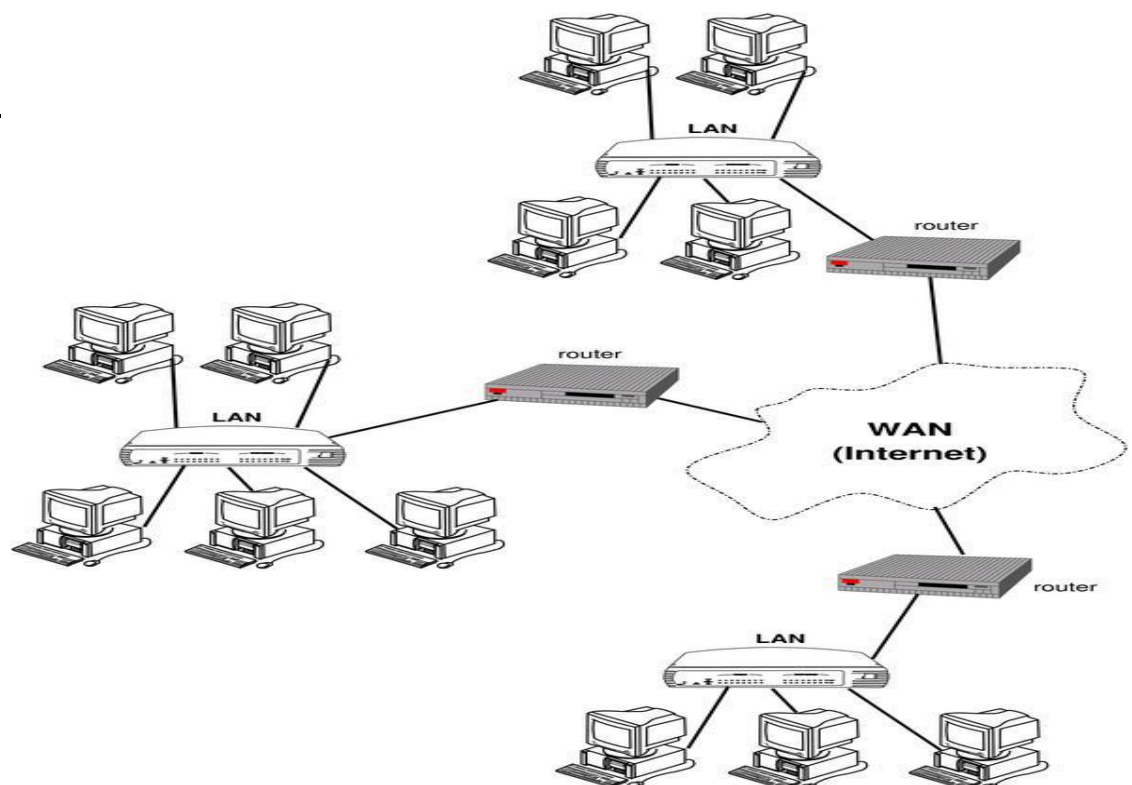
- High installation cost
- Higher delay (latency)
- Complex maintenance
- Security threats
- Depends on service providers

Applications

- Internet
- International banking
- Airline reservation systems
- Global companies
- Online education platforms

5. Internetwork

(Network of Networks)



Definition

An **Internetwork** is formed by connecting **two or more independent networks** using networking devices such as **routers**.

The Internet is the largest example of an internetwork.

Exam Definition

An internetwork is a collection of interconnected networks that communicate using routers and standard communication protocols.

Working of Internetwork

1. Different LANs and WANs exist independently.
2. Routers connect these separate networks.
3. Routers determine the best path.
4. Data travels across multiple networks until it reaches the destination.

Real-Life Example

Suppose:

- College LAN
- Bank WAN
- Home Wi-Fi
- Mobile Network

All communicate through the Internet.

This forms an **Internetwork**.

Advantages

- Global communication
- Scalability
- Resource sharing
- Better reliability
- Flexible network expansion

Disadvantages

- Complex configuration
- Higher security risks
- Costly management
- Routing complexity

Applications

- Internet
- Cloud Computing
- International companies
- Government networks
- E-commerce websites

Comparison Table (Exam View)

Feature	PAN	LAN	MAN	WAN	Internetwork
Full Form	Personal Area Network	Local Area Network	Metropolitan Area Network	Wide Area Network	Network of Networks
Coverage	1–10 m	Building/Campus	City	Country/World	Global
Speed	Medium	Very High	High	Medium to High	Depends on connected networks
Cost	Very Low	Low	Medium	High	Very High
Devices	Personal devices	Computers in one location	Multiple LANs	Multiple MANs/LANs	Multiple independent networks
Best Example	Bluetooth devices	College Lab	City Bank Branches	Internet	Internet connecting many networks

Easy Real-Life Memory Trick

- **PAN** → *My Devices* (Phone ↔ Smartwatch ↔ Earbuds)
- **LAN** → *One Building* (School Lab, Office)
- **MAN** → *One City* (Bank branches in Pune)
- **WAN** → *Whole Country/World* (International company)
- **Internetwork** → *Network of Networks* (The Internet)

➤ Network Software (BE Computer Engineering)

Network software defines **how computers communicate** with each other. It consists of **protocols, layers, services, and communication rules** that allow devices to exchange information correctly.

1. Protocol Hierarchies (Layered Architecture)

What is a Protocol?

A **protocol** is a set of rules and standards that govern how data is transmitted and received over a network.

Exam Definition

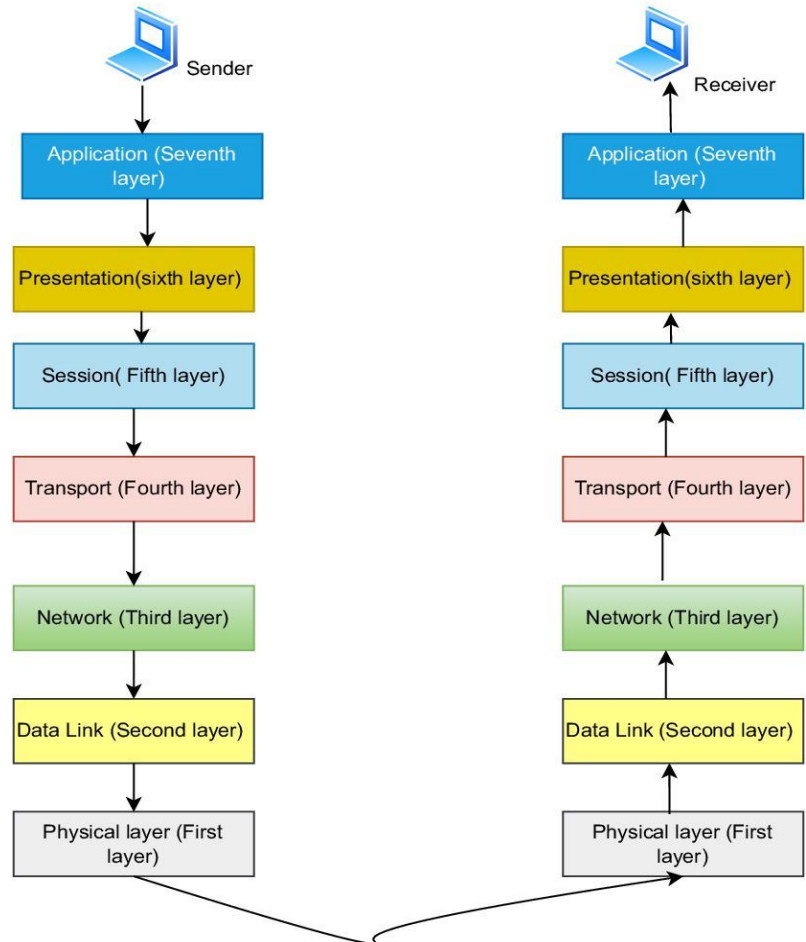
A protocol is a set of rules that defines how two or more devices communicate and exchange data over a computer network.

Real-Life Example

Imagine sending a letter.

Both the sender and receiver follow rules:

- Correct address
- Postal code
- Stamp
- Envelope



Similarly, computers follow protocols like **TCP/IP** to exchange data.

What is a Protocol Hierarchy?

Instead of performing all networking tasks together, communication is divided into **multiple layers**, where each layer performs a specific function.

Each layer:

- Provides services to the layer above.
- Uses services from the layer below.

Working

1. User sends data.
2. Data moves from the top layer to the bottom layer.
3. Each layer adds its own information (header).
4. Data travels through the network.
5. At the receiver, headers are removed layer by layer.
6. Original data reaches the application.

This process is called **Encapsulation and Decapsulation**.

Why Layering is Needed?

- Simpler design
- Easier troubleshooting
- Independent development
- Standardization
- Better interoperability

Advantages

- Modular design
- Easy maintenance
- Flexibility
- Standard communication
- Easy upgrades

Disadvantages

- Extra processing
- More overhead
- Slight communication delay

Real-Life Example

When opening a website:

1. Browser creates request.
2. Transport layer prepares segments.
3. Network layer adds IP address.
4. Data Link layer creates frames.
5. Physical layer sends bits.

Receiver performs the reverse process.

2. Design Issues for Layers

Each network layer has specific design responsibilities.

Major Design Issues

1. Addressing

Every device needs a unique address.

Example:

- IP Address
- MAC Address

Real-Life Example: Every house has a unique postal address.

2. Error Control

Detects and corrects transmission errors.

Methods:

- Checksum
- CRC
- Acknowledgement (ACK)

Example: If a file download is corrupted, missing packets are retransmitted.

3. Flow Control

Controls the rate of data transmission.

If the sender is faster than the receiver, the receiver's buffer may overflow.

Example: A teacher speaks slowly so students can understand.

4. Routing

Determines the best path for data.

Routers select the shortest or fastest route.

Example: Google Maps chooses the quickest route.

5. Multiplexing

Multiple users share one communication channel.

Example: Many people use the same mobile tower simultaneously.

6. Segmentation and Reassembly

Large messages are divided into smaller packets and later reassembled.

Example: A large book is packed into several boxes before shipping.

7. Reliability

Ensures data reaches the destination correctly.

Methods:

- ACK
- Retransmission
- Sequence numbers

8. Security

Protects data using:

- Encryption
- Authentication
- Firewalls

Advantages

- Reliable communication
- Error-free transmission
- Better security
- Efficient routing
- Improved performance

Disadvantages

- Complex implementation
- Increased processing
- More protocol overhead

3. Connection-Oriented Service

Definition

A **connection-oriented service** establishes a connection **before** sending data.

Data is transferred only after the connection is successfully established.

The best example is **TCP (Transmission Control Protocol)**.

Working

1. Sender requests connection.
2. Receiver accepts connection.
3. Data is transmitted.
4. Acknowledgements confirm receipt.
5. Connection is terminated.

Real-Life Example

Telephone Call

1. Dial number.
2. Receiver answers.
3. Conversation begins.
4. Call ends.

Similarly, TCP first establishes a connection and then transfers data.

Features

- Reliable
- Ordered delivery
- Error recovery
- Flow control

Advantages

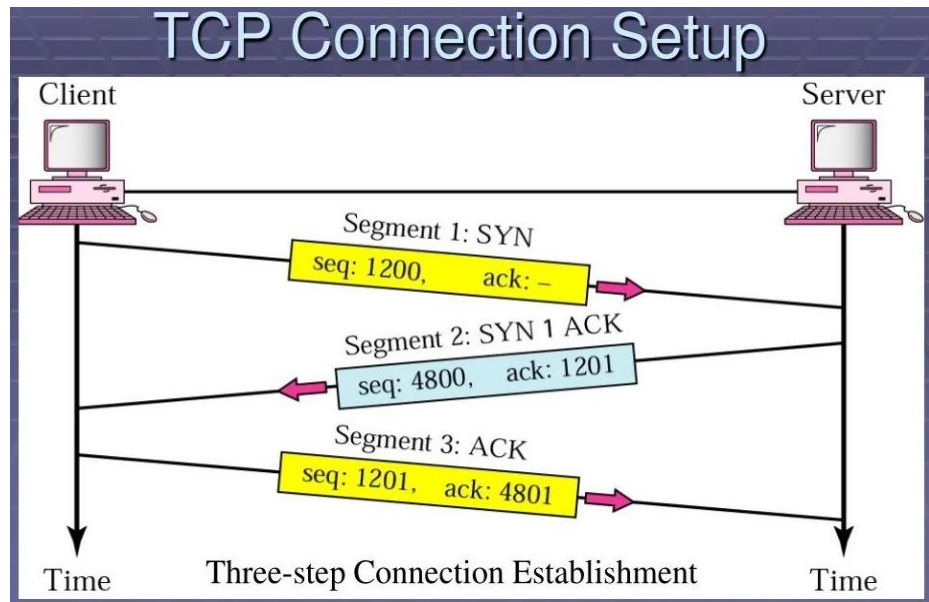
- Reliable communication
- No packet loss
- Ordered data delivery
- Error detection and correction

Disadvantages

- Slower due to connection setup
- Higher overhead
- More bandwidth usage

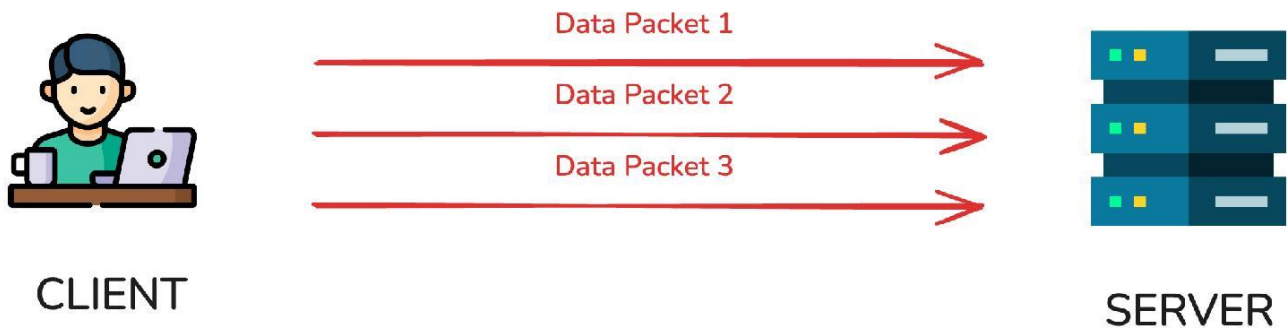
Applications

- Email
- Online Banking
- File Transfer (FTP)
- Web Browsing (HTTP over TCP)



4. Connectionless Service

UDP - Connectionless Protocol



- 🚀 Fast transmission - No handshake required
- ❌ No acknowledgments - Fire and forget approach
- 📦 Small header (8 bytes) - Minimal overhead
- ⚡ Low latency - Perfect for real-time applications

Definition

A **connectionless service** sends data **without establishing a connection**.

Each packet is independent.

The best example is **UDP (User Datagram Protocol)**.

Working

1. Sender sends packets immediately.
2. No connection setup.
3. No acknowledgements.
4. Receiver receives packets if available.

Real-Life Example

Sending a postcard.

You simply post it.

No confirmation is received whether it reaches the destination.

Features

- Fast
- Low overhead
- No guaranteed delivery
- No ordering

Advantages

- Very fast
- Low delay
- Simple implementation
- Less overhead

Disadvantages

- Packet loss possible
- No acknowledgements
- No retransmission
- Data may arrive out of order

Applications

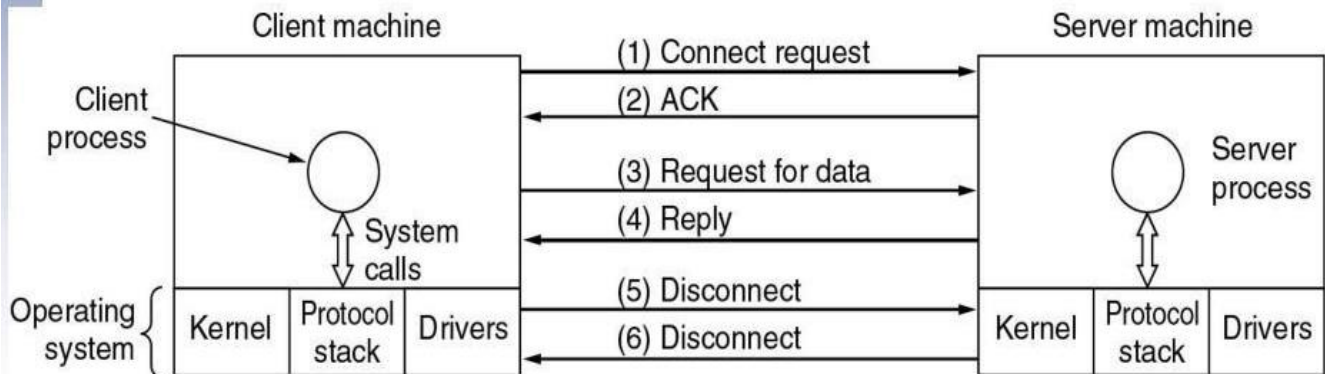
- Video Streaming
- Online Games
- Voice Calls (VoIP)
- Live Broadcast

Connection-Oriented vs Connectionless

Feature	Connection-Oriented	Connectionless
Connection Setup	Required	Not Required
Reliability	High	Low
Speed	Slower	Faster
Error Recovery	Yes	No
Protocol	TCP	UDP
Example	Phone Call	Postcard

5. Service Primitives

- **Packets sent in a simple client-server interaction on a connection-oriented network.**



Definition

Service primitives are basic operations used by one layer to request services from another layer. They define how communication starts, transfers data, and ends.

Main Service Primitives

1. LISTEN

The server waits for a connection request.

Example: Customer care waits for incoming calls.

2. CONNECT

Client requests connection.

Example: Dialing a phone number.

3. SEND

Transmits data.

Example: Sending a WhatsApp message.

4. RECEIVE

Accepts incoming data.

Example: Receiving an email.

5. DISCONNECT

Terminates communication.

Example: Hanging up a phone call.

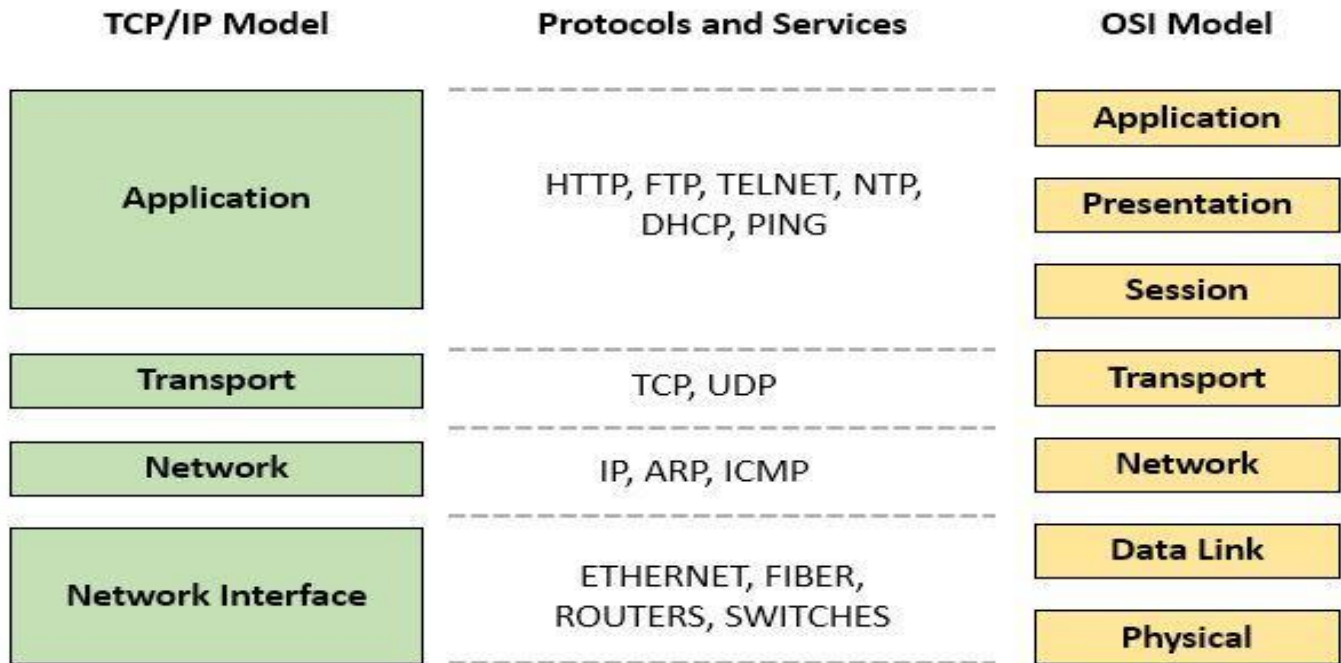
Advantages

- Organized communication
- Easy implementation
- Reliable interaction
- Standardized operations

Disadvantages

- Slight overhead
- More processing for each operation

6. Relationship Between Services and Protocols



Service

A **service** is **what a layer provides** to the layer above it.

It describes **what is offered**, not how it is implemented.

Example

The transport layer provides reliable delivery to applications.

Protocol

A **protocol** defines **how two corresponding layers communicate**.

It specifies:

- Message format
- Timing
- Error handling
- Data exchange rules

Relationship

- **Service** = What is provided.
- **Protocol** = How the service is implemented.

Protocols implement the services offered by each layer.

Real-Life Example

Restaurant Analogy:

- **Service:** Food is served to customers.
- **Protocol:** The rules followed by waiters and kitchen staff to prepare and deliver the food.

Customers care about the service, not the internal process.

Advantages

- Clear separation of responsibilities
- Independent protocol changes
- Easier maintenance
- Better flexibility

Disadvantages

- More abstraction
- Additional protocol overhead

Exam Difference: Service vs Protocol

Service	Protocol
Describes what is provided	Describes how communication occurs
Interface between layers	Rules between peer layers
Visible to upper layer	Used internally by corresponding layers
Independent of implementation	Defines implementation details
Example: Reliable file transfer	Example: TCP

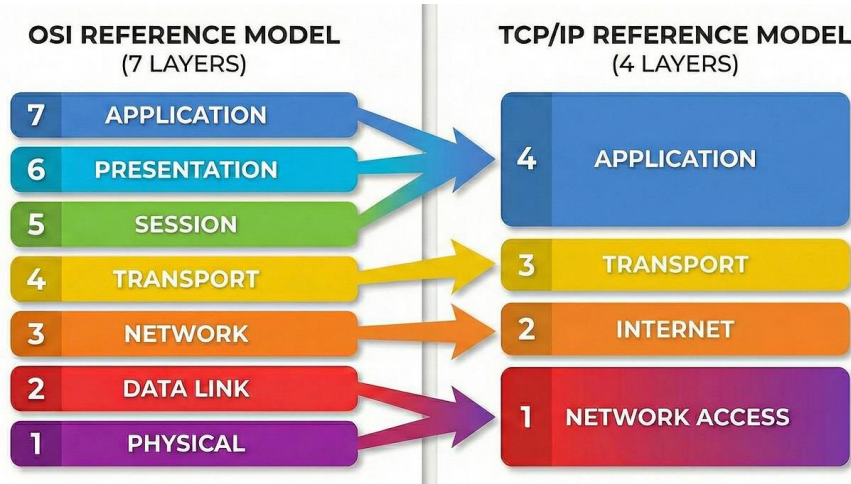
One-Line Revision (Very Important)

- **Protocol** → Rules for communication.
- **Protocol Hierarchy** → Communication divided into layers.
- **Design Issues** → Addressing, routing, flow control, error control, security, reliability, multiplexing.
- **Connection-Oriented** → Connection first, then data (TCP, phone call).
- **Connectionless** → Send data directly (UDP, postcard).
- **Service Primitives** → LISTEN, CONNECT, SEND, RECEIVE, DISCONNECT.
- **Service** → What a layer offers.
- **Protocol** → How the service is implemented.

➤ Unit I: Reference Models – OSI Model and TCP/IP Model (BE Computer Engineering)

Reference models provide a **standard framework** for understanding how data is transmitted over a computer network. Instead of handling all communication in one step, networking tasks are divided into layers.

What is a Reference Model?



Definition

A **Reference Model** is a conceptual framework that divides the communication process into multiple layers. Each layer performs a specific function and communicates with its adjacent layers.

Exam Definition

A reference model is a layered framework that standardizes the communication process between network devices by dividing networking tasks into separate layers.

Why are Reference Models Needed?

Without layers:

- Communication becomes complex.
- Troubleshooting is difficult.
- Devices from different manufacturers may not communicate.

With layers:

- Standardization
- Easy maintenance
- Better interoperability
- Modular design

Real-Life Example

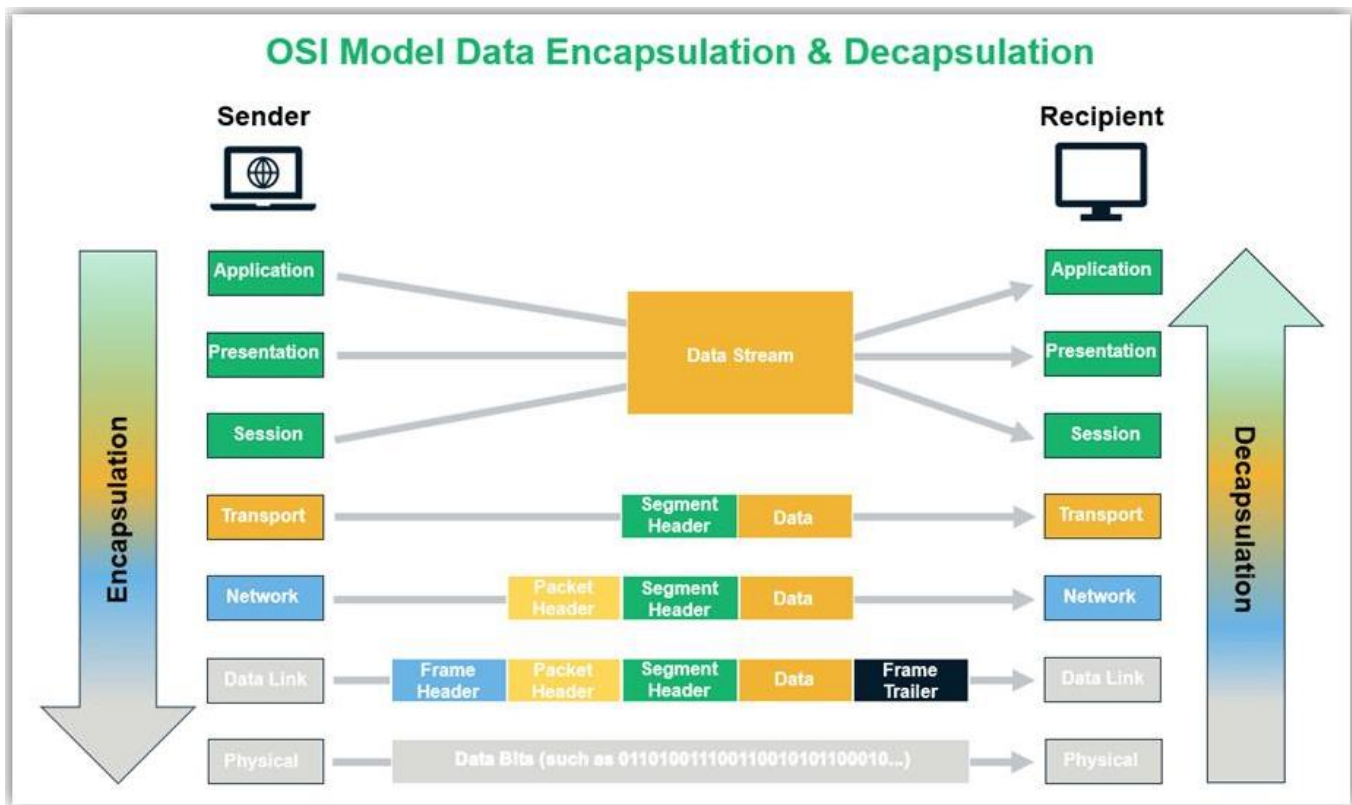
Think of **sending a courier parcel**.

Different departments perform different jobs:

1. Customer Service accepts the parcel.
2. Packing Department packs it.
3. Transportation moves it.
4. Local Delivery delivers it.
5. Receiver gets the parcel.

Similarly, networking is divided into layers.

- **OSI Reference Model**



Introduction

The **OSI (Open Systems Interconnection) Reference Model** was developed by the International Organization for Standardization.

It contains **7 layers**, where each layer performs a specific networking task.

Exam Definition

The OSI model is a seven-layer reference model developed by ISO to standardize network communication between different computer systems.

Seven Layers of OSI Model

Layer Number Layer Name Main Function

7	Application	User interface and network services
6	Presentation	Data translation, encryption, compression
5	Session	Session establishment and management
4	Transport	Reliable delivery and error recovery
3	Network	Routing and logical addressing
2	Data Link	Framing, MAC addressing, error detection
1	Physical	Transmission of bits over media

Layer 7 – Application Layer

Function

Provides services directly to the user applications.

Examples:

- Email
- Web Browsing
- File Transfer

Protocols

- HTTP
- HTTPS
- FTP
- SMTP
- DNS

Real-Life Example

Opening a web browser to visit a website.

Layer 6 – Presentation Layer

Function

Responsible for:

- Data Translation
- Encryption
- Compression

Example

Before sending a password:

- Encrypt it
- Compress multimedia data

Real-Life Example

A PDF file created on Windows can be opened on Linux because this layer handles data format compatibility.

Layer 5 – Session Layer

Function

Establishes, maintains, and terminates communication sessions.

Real-Life Example

During a Google Meet call:

- Starts the session
 - Keeps it active
 - Ends it when the meeting finishes
-

Layer 4 – Transport Layer

Function

Provides:

- Reliable communication
- Error recovery
- Flow control
- Segmentation

Protocols

- TCP , UDP

Real-Life Example

Downloading a file:

If packets are lost, TCP requests retransmission.

Layer 3 – Network Layer

Function

Responsible for:

- Logical addressing
- Routing
- Path selection

Protocol

IP

Device

Router

Real-Life Example

Google Maps finds the best route to your destination.

Similarly, routers find the best path for packets.

Layer 2 – Data Link Layer

Function

Responsible for:

- Framing
- MAC Address
- Error detection
- Access to transmission media

Device

Switch

Real-Life Example

Every apartment has a flat number.

Similarly, every network card has a unique MAC address.

Layer 1 – Physical Layer

Function

Transmits raw bits using:

- Ethernet Cable
- Fiber Optics
- Radio Waves

Devices

- Hub
- Repeater
- Cable

Real-Life Example

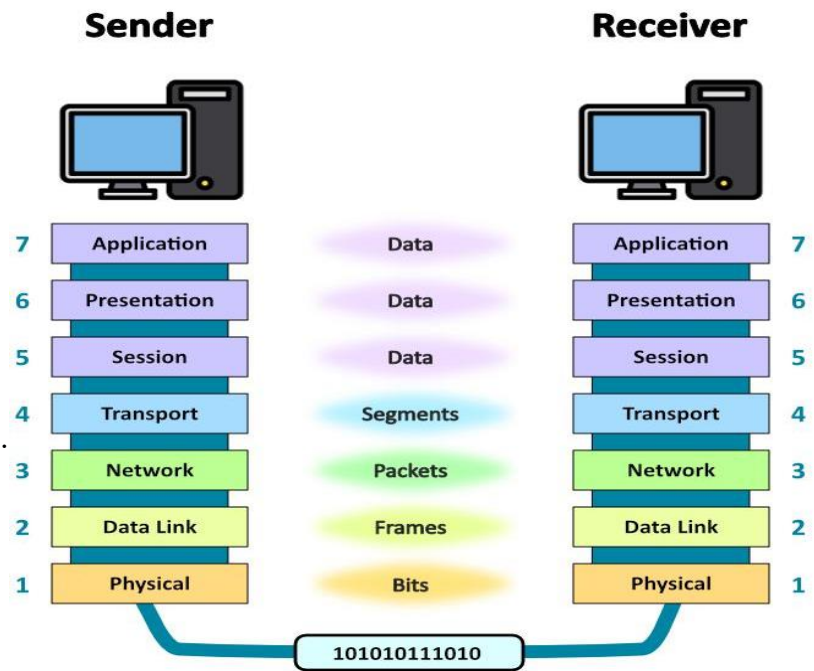
Electric wires carry electricity.

Similarly, cables carry binary bits.

Working of OSI Model :-

Step-by-Step

1. User writes an email.
2. Application Layer prepares data.
3. Presentation Layer encrypts it.
4. Session Layer creates a session.
5. Transport Layer divides data into segments.
6. Network Layer adds IP address.
7. Data Link Layer creates frames.
8. Physical Layer converts frames into bits.
9. Bits travel through the network.
10. Receiver removes headers in reverse order (Decapsulation).
11. Original message reaches the application.



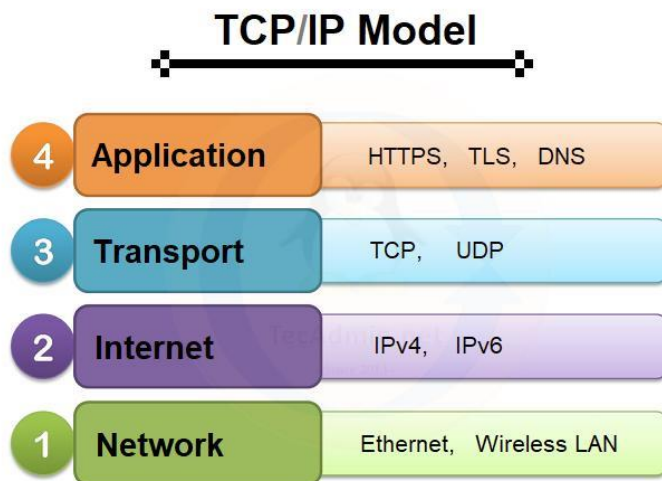
Advantages of OSI Model

- Standardized architecture
- Easy troubleshooting
- Modular design
- Easy protocol replacement
- Vendor independence
- Better learning model

Disadvantages of OSI Model

- Complex implementation
 - Seven layers increase overhead
 - Mostly theoretical
 - Rarely implemented exactly as defined
-

- **TCP/IP Reference Model :-**



Introduction

The **TCP/IP (Transmission Control Protocol/Internet Protocol)** model is the practical networking model used by the Internet.

It has **4 layers**.

Exam Definition

TCP/IP is a four-layer reference model that defines the communication architecture used on the Internet.

Four Layers of TCP/IP Model

Layer	Function
Application	User services
Transport	End-to-end communication
Internet	Routing and IP addressing
Network Access	Physical transmission and framing

1. Application Layer

Functions

Combines the functions of:

- OSI Application
- Presentation
- Session

Protocols:

- HTTP ,HTTPS
- FTP
- SMTP
- DNS

Example

Opening YouTube.

2. Transport Layer

Functions

- Reliable communication
- Flow control
- Error recovery

Protocols:

- TCP
- UDP

Example

Downloading software using TCP.

3. Internet Layer

Functions

- Routing
- Logical addressing
- Packet forwarding

Protocol:

- IP

Device

Router

Example

Finding the best Internet route.

4. Network Access Layer

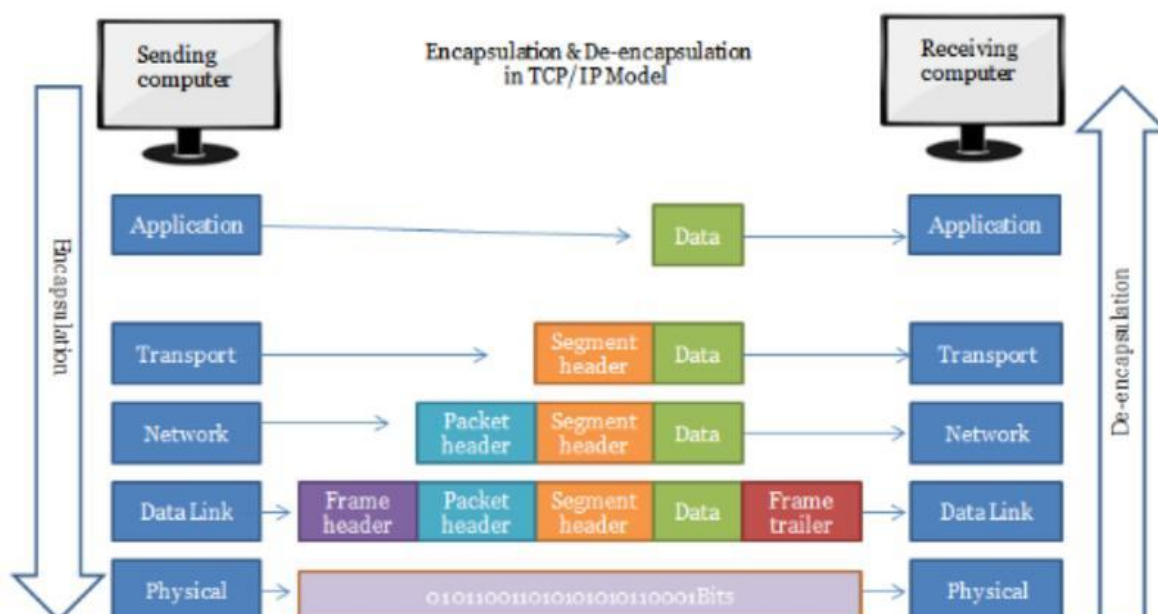
Functions

- Framing
- Physical transmission
- MAC Address

Example

Ethernet communication.

Working of TCP/IP Model



Steps

1. User opens browser.
2. Application creates HTTP request.
3. Transport Layer adds TCP header.
4. Internet Layer adds IP header.
5. Network Access Layer converts packets into frames and bits.
6. Data travels.
7. Receiver performs reverse processing.

Advantages of TCP/IP

- Practical implementation
- Used worldwide
- Highly scalable
- Reliable
- Supports heterogeneous networks
- Internet standard

Disadvantages of TCP/IP

- Less strict layer separation
- More difficult to replace protocols
- Limited security in the basic model
- Not ideal for studying networking concepts

OSI Model vs TCP/IP Model

Feature	OSI Model	TCP/IP Model
Developed By	ISO	DARPA
Number of Layers	7	4
Nature	Reference model	Practical protocol suite
Use	Teaching and design	Real Internet communication
Session Layer	Separate	Included in Application
Presentation Layer	Separate	Included in Application
Network Layer	Network Layer	Internet Layer
Physical Layer	Separate	Part of Network Access
Protocol Dependency	Protocol independent	Based on TCP/IP protocols

Mapping of OSI and TCP/IP Models

OSI Layer TCP/IP Layer

Application Application

Presentation Application

Session Application

Transport Transport

Network Internet

Data Link Network Access

Physical Network Access

Real-Life Example (Complete Communication)

Suppose you open **www.google.com**:

1. **Application Layer** – Browser creates HTTP request.
2. **Transport Layer** – TCP divides data into segments.
3. **Internet Layer** – IP adds source and destination IP addresses.
4. **Network Access Layer** – Data is converted into frames and bits.
5. Data travels through routers.
6. Google's server receives the request.
7. Response returns using the same layered process.

Memory Trick for OSI Layers

Top to Bottom

A – Application

P – Presentation

S – Session

T – Transport

N – Network

D – Data Link

P – Physical

Mnemonic:

All People Seem To Need Data Processing

Memory Trick for TCP/IP Layers

A – T – I – N

- **A** – Application
- **T** – Transport
- **I** – Internet
- **N** – Network Access

Exam Points (Very Important)

Definitions to Remember

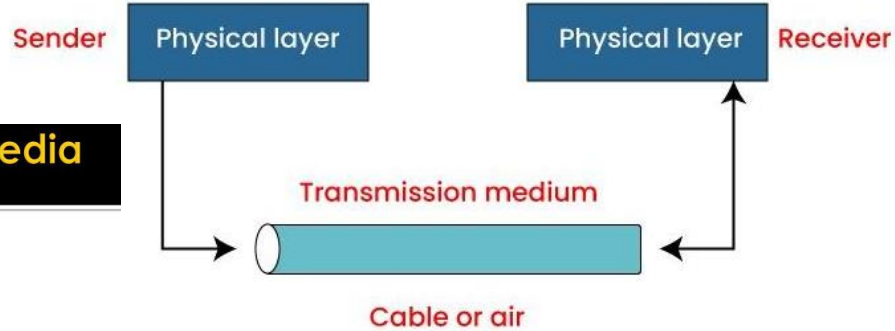
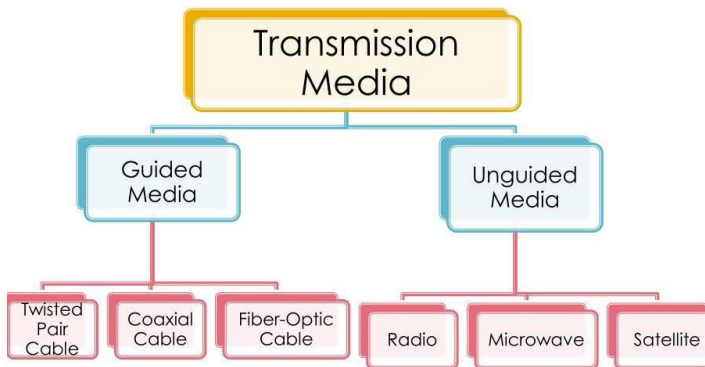
- **OSI Model:** A 7-layer reference model developed by ISO for standardizing network communication.
- **TCP/IP Model:** A 4-layer protocol model used for Internet communication.

➤ Physical Layer

The **Physical Layer** is the **first (lowest) layer** of the OSI model. It is responsible for the **actual transmission of raw binary bits (0s and 1s)** over a communication medium.

1. Physical Layer

Physical Layer Transmission Media



Definition

The **Physical Layer** transmits **raw bits (0s and 1s)** from one device to another using electrical, optical, or radio signals.

It defines:

- Transmission media
- Signal type
- Voltage levels
- Data rate
- Connectors
- Physical topology

Exam Definition

The **Physical Layer** is the **first layer of the OSI model** responsible for transmitting raw bits over a **physical communication medium**.

Main Functions of the Physical Layer

- Transmits raw bits (0 and 1)
- Converts data into electrical, optical, or radio signals
- Defines transmission media
- Specifies connectors and cables
- Determines data transmission speed
- Controls signal synchronization
- Defines physical topology

Working of the Physical Layer

1. The Data Link Layer sends a frame.
2. The Physical Layer converts the frame into **bits**.
3. Bits are converted into signals.
4. Signals travel through cables or wireless media.
5. The receiver converts signals back into bits.
6. Bits are passed to the Data Link Layer.

Real-Life Example

Suppose you send a photo through WhatsApp.

- Application prepares the photo.
- Transport divides it into packets.
- Network adds IP addresses.
- Data Link creates frames.
- **Physical Layer sends electrical or radio signals** through Wi-Fi or mobile networks.

Advantages

- High-speed transmission
- Simple implementation
- Supports wired and wireless communication
- Standardized hardware interfaces

Disadvantages

- No error detection
 - No security
 - Only transmits bits
 - Cannot identify the destination
-

2. Guided Transmission Media

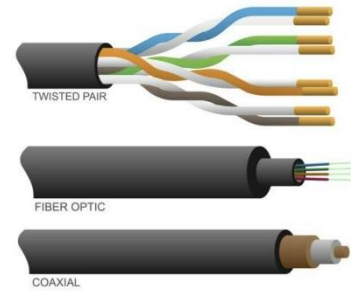
(Guided media is also called **Wired Media**.)

Definition

Guided transmission media uses **physical cables** to transmit data from sender to receiver.

Types

1. Twisted Pair Cable
2. Coaxial Cable
3. Optical Fiber Cable



A. Twisted Pair Cable

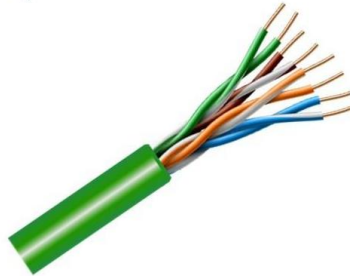
Definition

A **Twisted Pair Cable** consists of **two insulated copper wires twisted together** to reduce electromagnetic interference.

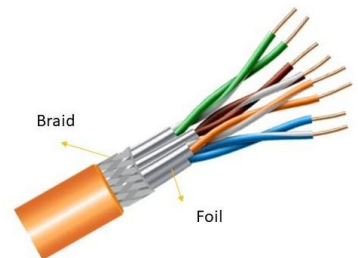
Types

- UTP (Unshielded Twisted Pair)
- STP (Shielded Twisted Pair)

Unshielded Twisted Pair (UTP)



Shielded Twisted Pair (STP)



Working

1. Data is converted into electrical signals.
2. Signals travel through copper wires.
3. Twisting reduces noise and interference.
4. The receiver converts the signals back into data.

Real-Life Example

Ethernet cable connecting:

- Computer ↔ Switch
- Router ↔ Laptop

Advantages

- Low cost
- Easy installation
- Flexible
- Widely available

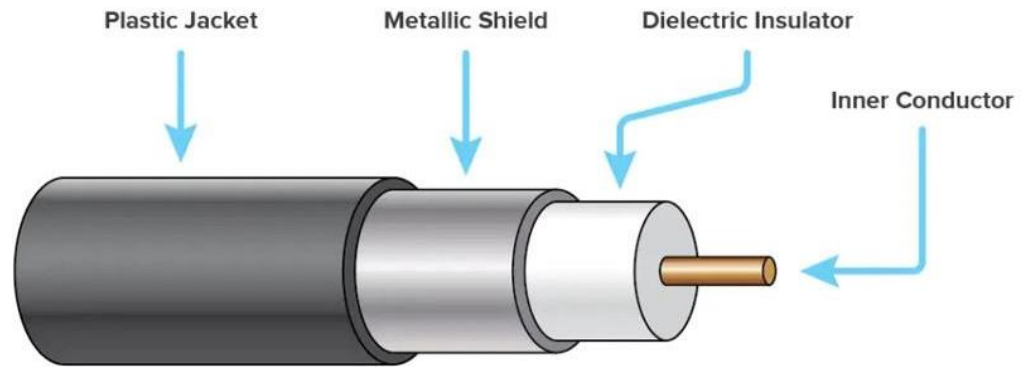
Disadvantages

- Limited distance (about 100 m)
- Lower bandwidth
- Affected by electromagnetic interference

Applications

- LAN
- Telephone lines
- Office networks

B. Coaxial Cable



Definition

A **Coaxial Cable** has a central copper conductor surrounded by insulation, metallic shielding, and an outer protective jacket.

Working

Electrical signals travel through the central conductor, while the metallic shield protects against interference.

Real-Life Example

- Cable TV
- CCTV cameras
- Broadband Internet

Advantages

- Better noise protection
- Longer distance than twisted pair
- Higher bandwidth

Disadvantages

- More expensive
- Difficult installation
- Less flexible

Applications

- Television
- Cable Internet
- Security systems

C. Optical Fiber Cable

Definition

An **Optical Fiber Cable** transmits data as **light pulses** through thin strands of glass or plastic.

Working

1. Data is converted into light.
2. Light travels through the fiber core.
3. Total Internal Reflection keeps the light inside the fiber.
4. The receiver converts the light back into electrical signals.

Real-Life Example

- High-speed broadband
- Undersea Internet cables
- 5G backbone
- Data centers

Advantages

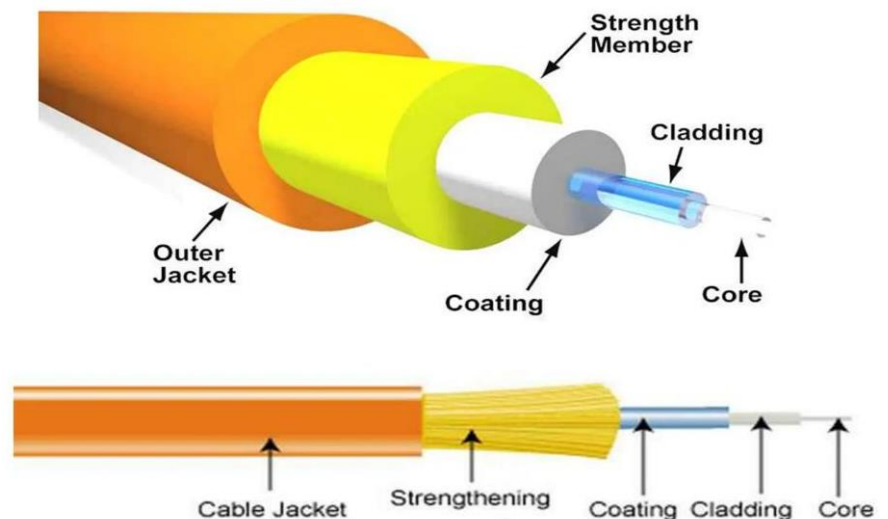
- Very high speed
- Long-distance communication
- Immune to electromagnetic interference
- High security
- Large bandwidth

Disadvantages

- Expensive
- Fragile
- Difficult installation and repair

Applications

- Internet backbone
- Hospitals
- Banks
- Telecommunications



Comparison of Guided Media

Feature	Twisted Pair	Coaxial Cable	Optical Fiber
Signal	Electrical	Electrical	Light
Speed	Medium	High	Very High
Cost	Low	Medium	High
Distance	Short	Medium	Very Long
EMI Resistance	Low	Medium	Excellent
Application	LAN	Cable TV	High-speed Internet

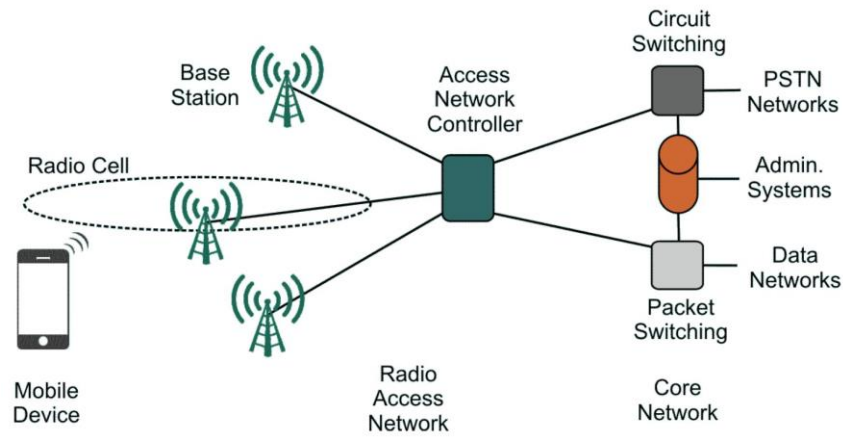
3. Wireless Transmission (Unguided Media)

Definition

Wireless transmission sends data through **electromagnetic waves** instead of cables.

Types

- Radio Waves
- Microwaves
- Infrared
- Satellite Communication



A. Radio Waves

Working

Signals are transmitted in all directions.

Example

- FM Radio
- Bluetooth
- Wi-Fi

Advantages

- Long range
- Low cost
- Easy installation

Disadvantages

- Interference
- Less secure
- Limited bandwidth

B. Microwaves

Working

Microwave antennas communicate in a straight line (Line-of-Sight).

Example

- Mobile towers
- TV transmission

Advantages

- High speed
- Long distance

Disadvantages

- Requires line-of-sight
- Weather affects communication

C. Infrared

Working

Uses infrared light for short-range communication.

Example

- TV Remote
- Remote-controlled toys

Advantages

- Secure
- Low interference

Disadvantages

- Very short range
- Cannot pass through walls

D. Satellite Communication

Working

1. Ground station sends signals to a satellite.
2. The satellite receives and amplifies the signals.
3. The satellite retransmits them to another ground station.

Real-Life Example

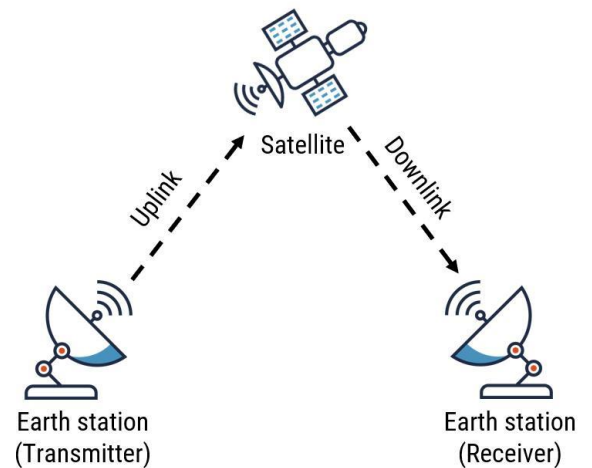
- GPS
- DTH Television
- Weather forecasting
- International communication

Advantages

- Global coverage
- Long-distance communication
- Supports remote areas

Disadvantages

- High cost
- Signal delay (latency)
- Weather effects



Comparison of Wireless Media

Medium	Range	Speed	Example
Radio Waves	Long	Medium	Wi-Fi, Bluetooth
Microwave	Very Long	High	Mobile Towers
Infrared	Very Short	Medium	TV Remote
Satellite	Global	High	GPS, DTH

4. Telephone System

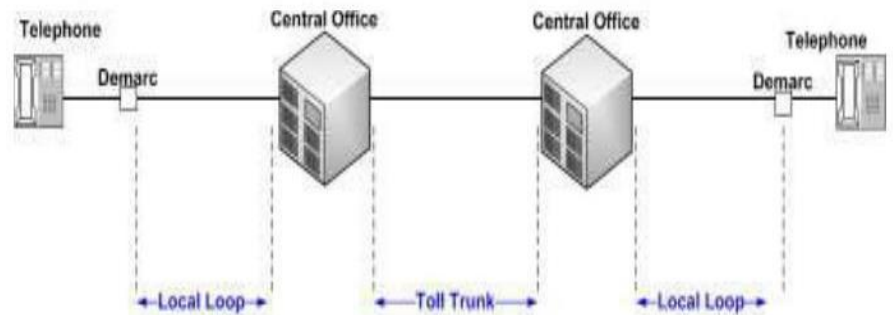
Local loop and Trunk lines

Definition

A **Telephone System** is a communication network that allows voice and data transmission between users over wired or wireless networks.

Modern telephone systems support:

- Voice calls
- Internet
- Video calls
- Fax



Components of Telephone System

1. Telephone Device
2. Local Loop
3. Telephone Exchange
4. Trunk Lines
5. Switching System

Working of Telephone System

1. User dials a phone number.
2. The call travels through the **local loop** to the telephone exchange.
3. The exchange identifies the destination.
4. If local, the call is connected directly.
5. If remote, trunk lines carry the call to another exchange.
6. The destination telephone rings.
7. Communication begins.

Real-Life Example

Suppose you call your friend in another city.

- Your phone → Local Exchange
- Local Exchange → Trunk Line
- Trunk Line → Friend's Exchange
- Friend's Exchange → Friend's Phone

Types of Telephone Systems

1. PSTN (Public Switched Telephone Network)

Uses circuit switching for traditional landline communication.

2. Mobile Telephone Network

Uses cellular towers and wireless communication.

3. VoIP (Voice over Internet Protocol)

Uses the Internet to make voice calls.

Examples:

- WhatsApp Calls
- Skype
- Zoom

Advantages

- Fast communication
- Reliable
- Global connectivity
- Supports voice and data
- Emergency communication

Disadvantages

- Infrastructure cost
 - Network congestion
 - Signal interference (wireless)
 - Security concerns
-

Guided vs Wireless Transmission

Feature	Guided Media	Wireless Media
Medium	Cable	Air
Cost	Medium	Low to Medium
Speed	High	Medium to High
Security	High	Lower
Installation	Difficult	Easy
Mobility	No	Yes
Examples	Ethernet, Fiber	Wi-Fi, 5G, Bluetooth

Memory Tricks (Exam)

Guided Media

TCO

- **T** – Twisted Pair , **C** – Coaxial Cable , **O** – Optical Fiber

Wireless Media

RMIS

- **R** – Radio , **M** – Microwave , **I** – Infrared, **S** – Satellite
-

Last-Minute Revision Points

- **Physical Layer:** Transmits raw bits using electrical, optical, or radio signals.
- **Guided Media:** Twisted Pair, Coaxial Cable, Optical Fiber.
- **Wireless Media:** Radio Waves, Microwaves, Infrared, Satellite.
- **Optical Fiber:** Highest speed, longest distance, uses light.
- **Twisted Pair:** Cheapest and common in LANs.
- **Coaxial Cable:** Used for cable TV and broadband.
- **Telephone System:** Phone → Local Loop → Exchange → Trunk Line → Destination.
- **PSTN** uses circuit switching, while **VoIP** uses the Internet.

➤ Unit I – Narrowband and Broadband Communication Systems

Communication systems are classified based on the **bandwidth** they use. The two major types are:

1. **Narrowband Communication System**
2. **Broadband Communication System**

What is Bandwidth?

Definition

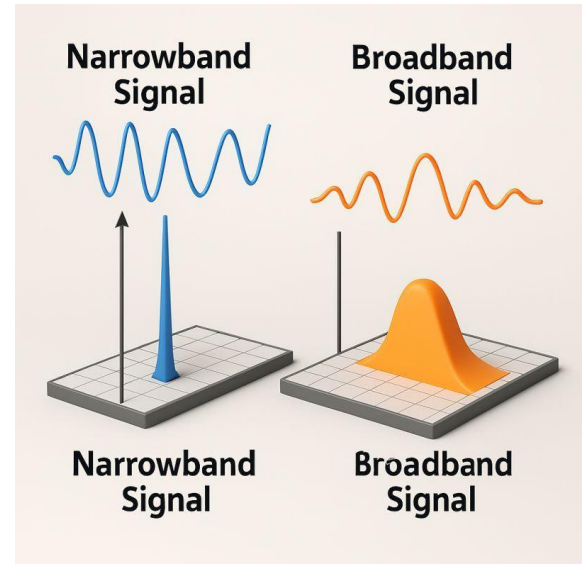
Bandwidth is the **range of frequencies** that a communication channel can carry.

It is measured in **Hertz (Hz)**.

- Higher bandwidth = More data transmitted per second.
- Lower bandwidth = Less data transmitted per second.

Exam Definition

Bandwidth is the difference between the highest and lowest frequencies that can be transmitted through a communication channel.



Real-Life Example

Imagine a highway.

- **Narrow road (1 lane)** → Few vehicles can travel at a time (**Narrowband**).
- **Wide highway (6 lanes)** → Many vehicles can travel simultaneously (**Broadband**).

Similarly:

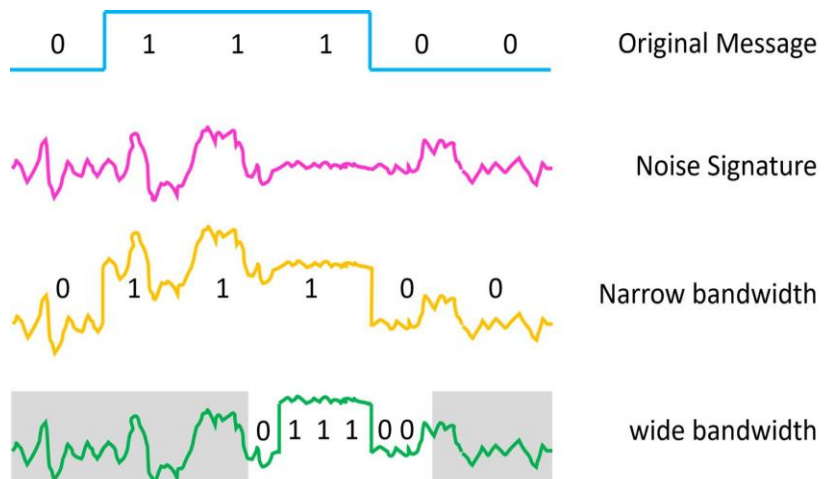
- Narrow bandwidth carries less data.
- Broadband carries much more data.

1. Narrowband Communication System

Definition

A **Narrowband Communication System** uses a **small range of frequencies (low bandwidth)** to transmit data.

It is mainly designed for **voice communication** and **low-speed data transmission**.



Exam Definition

A narrowband communication system uses a small bandwidth to transmit low-speed data and voice signals.

Working of Narrowband Communication

1. Sender generates data or voice.
2. Data is converted into electrical signals.
3. Signals are transmitted over a low-bandwidth channel.
4. Receiver receives and converts the signals back into data or voice.

Because only a small frequency range is available:

- Data transfer is slower.
- Fewer signals can be transmitted simultaneously.

Characteristics

- Small bandwidth
- Low data transfer rate
- Mostly analog communication
- Supports voice better than multimedia
- Lower equipment cost

Real-Life Example

Example 1: Traditional Landline Telephone

When you make a call using an old telephone:

- Only your voice is transmitted.
- The telephone channel uses approximately **300 Hz to 3400 Hz**.

This is a **narrowband system**.

Example 2: Dial-up Internet

Older Internet connections using telephone lines provided speeds of about **56 kbps**, making them narrowband.

Applications

- Landline telephones
- Voice communication
- Walkie-talkies
- Fax machines
- Pager systems

Advantages

- Low installation cost
- Simple technology
- Low power consumption
- Reliable for voice communication
- Easy maintenance

Disadvantages

- Very slow data transmission
- Cannot support HD video
- Unsuitable for large file transfers
- Limited bandwidth
- Poor multimedia performance

2. Broadband Communication System

Definition

A Broadband Communication System

uses a **large range of frequencies** (**high bandwidth**) to transmit large amounts of data at high speed.

It supports:

- Voice
- Video
- Images
- Internet
- Streaming
- Cloud applications

Exam Definition

A broadband communication system uses a wide bandwidth to provide high-speed transmission of voice, video, and data simultaneously.

Working of Broadband Communication

1. User requests data (e.g., opens YouTube).
2. Data is divided into packets.
3. Packets travel through high-bandwidth media (fiber optic, cable, DSL, or wireless broadband).
4. The receiver receives packets at high speed.
5. Video or data is reconstructed for the user.

Characteristics

- High bandwidth
- High-speed communication
- Supports multimedia
- Simultaneous services
- Digital communication

Real-Life Example

Example 1: Fiber Internet

When watching a **4K YouTube video**, broadband provides enough bandwidth for smooth playback.

Example 2: Home Wi-Fi

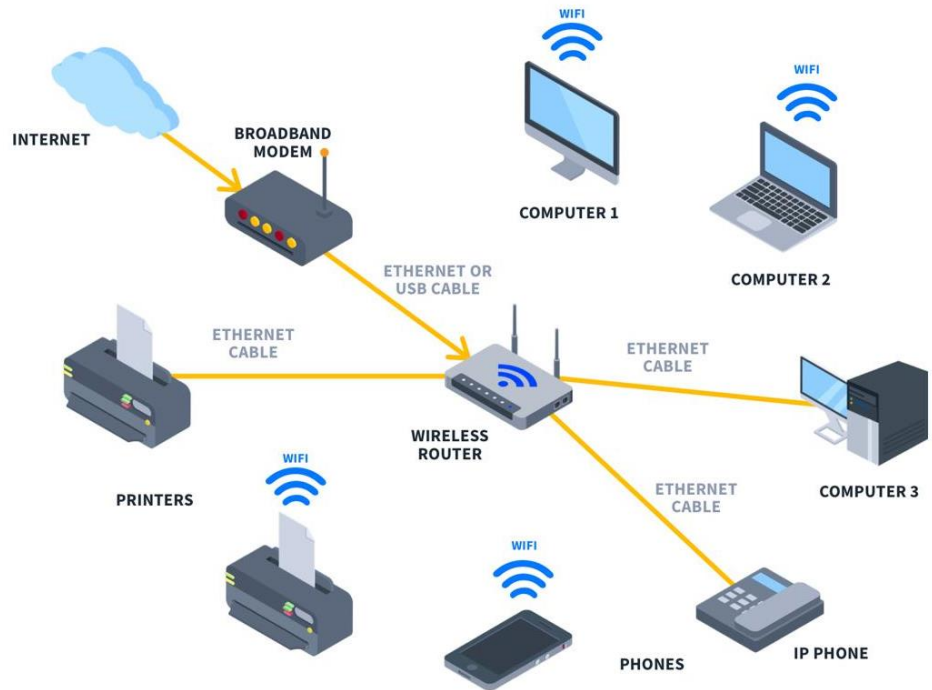
Your family can:

- Watch Netflix
- Attend Zoom meetings
- Play online games
- Download files

All at the same time using one broadband connection.

Applications

- Fiber-optic Internet
- Wi-Fi
- Cable Internet
- 4G/5G mobile networks



- Online gaming
- Video conferencing
- IPTV
- Cloud computing

Advantages

- Very high speed
- Supports HD/4K streaming
- Simultaneous voice, video, and data
- Reliable Internet access
- Supports many users

Disadvantages

- Higher installation cost
- More expensive equipment
- Requires advanced infrastructure
- Performance may reduce with network congestion

Comparison of Narrowband and Broadband

Feature	Narrowband	Broadband
Bandwidth	Small	Large
Data Speed	Low	High
Data Capacity	Small	Large
Signal Type	Mostly Voice	Voice, Video, Data
Cost	Low	Higher
Multimedia Support	No	Yes
Internet Speed	Slow	Fast
Applications	Telephone, Fax Fiber, Wi-Fi, 5G	

Working Comparison

Narrowband

Sender



Small Bandwidth Channel



Receiver

- One signal at a time
- Slow communication

Broadband

Voice

\

Video ----> High Bandwidth Channel ----> Receiver

/

Data

- Multiple signals simultaneously
- High-speed communication

Real-Life Comparison

Narrowband Example

A village uses an old landline telephone.

- Voice call quality is good.
- Internet speed is very low.
- No HD video streaming.

Broadband Example

A modern home has fiber broadband.

Family members can simultaneously:

- Attend an online class
- Watch Netflix
- Download software
- Play online games

without significant interruption.

Advantages and Disadvantages Summary

Narrowband Advantages

Low cost

Easy installation

Reliable voice communication

Low power

Broadband Advantages

Very high speed

Supports multimedia

Multiple users simultaneously

Cloud and streaming support

Narrowband Disadvantages

Slow speed

Limited bandwidth

No HD streaming

Not suitable for large files

Broadband Disadvantages

High installation cost

Expensive infrastructure

Network congestion may affect speed

Requires modern equipment
